

Maryam Sadeghabadi

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RESEARCH INTERESTS

- Reliable Medical Image Analysis
- Uncertainty Quantification
- Spurious Correlation
- LLM Reasoning

EDUCATION

M.S. in Computer Engineering, Bioinformatics

Sharif University of Technology

Sep 2024-Ongoing

Tehran, Iran

- Supervisor: Dr. M. H. Rohban
- GPA: 17.5/20

B.S. in Computer Science

Amirkabir University of Technology (Tehran Polytechnic)

2020-2024

Tehran, Iran

- GPA: 18.84/20
- Ranked in the top 10 among 90 students

RESEARCH EXPERIENCE

Graduate Research Assistant

Sharif University of Technology

Robust and Interpretable Machine Learning lab (RIML lab)

- Investigating uncertainty quantification and deep generative modeling techniques for radiological image analysis and prediction. Designing and evaluating patient-specific prediction pipelines with rigorous validation methodologies on medical imaging datasets. Developing robust AI frameworks for clinically relevant decision support applications.

PROJECTS

LLM-Based Scientific Reasoning and Knowledge Generation

Fall 2025

Research Intern - RIML lab

- Explored the application of large language models (LLMs) to scientific discovery workflows through literature review and experimental investigation. Evaluated LLM-based approaches for hypothesis generation, knowledge extraction, and automated scientific reasoning.

Automated Spurious Feature Discovery in Image Classifiers

Fall 2025

Deep Generative Models Project

- Conducted research on spurious correlation discovery and bias auditing in image classification models. Designed and evaluated data-driven methodologies for uncovering latent sources of model bias and assessing their impact on predictive performance.

Deep Generative Models

Fall 2025

Coursework Portfolio

- Implemented and analyzed autoregressive models for audio and image generation; derived theoretical properties of Gaussian AR models and Real NADE.
- Built VAE variants from scratch for probabilistic graph forecasting and conditional image generation; studied posterior collapse and Cauchy-Schwarz divergence.
- Constructed normalizing flows and continuous normalizing flows; applied Glow to image inpainting.

- Implemented score-based/energy-based models with score-matching variants (ESM, ISM, DSM) and MCMC sampling.

Document Retrieval with BIM, BM25, N-gram language models and Bert Fall 2023
Information Retrieval Course

- Developed N-gram models for document retrieval using Mercer-Jelinek smoothing and implemented semantic vector representations with BERT and word2vec for improved query and document analysis.

Presented Recent Advancements on MultiModal IR based on Surveys Fall 2023
Survey Presentation

- Presented a comprehensive survey on recent advancements in Multimodal Information IR, focusing on image retrieval techniques,highlighting key developments and methodologies across recent literature.

HONORS AND AWARDS

- Ranked **5th** among 88 students at the University of Amirkabir(GPA = 19.1/20) Spring 2022
- Eligible to pursue a **second major** as a result of exceptional academic performance Spring 2021

RELATED COURSES

Intelligent Analysis of Biomedical Images *	20/20	Computational Data Mining *	19/20
Drug Discovery *	19/20	Combinatorial Optimization	19.8/20
Deep Generative Models *	16.4/20	Artificial Intelligence	20/20
Data Mining	20/20	Linear Optimization	20/20

* Indicates graduate courses

TEACHING EXPERIENCE

Spring 2024	DATA MINING	Dr. Shakeri
Spring 2024	NEURAL NETWORKS AND DEEP LEARNING	Dr. Shakeri
Spring 2024	MACHINE LEARNING	Dr. Motahari
Fall 2023	ARTIFICIAL INTELLIGENCE	Dr. Ghatee

VOLUNTEER EXPERIENCE

Scientific Team Leader Mar2021-Aug2022
Computer Webinar Series(CWS)

- Led content planning, speaker coordination, and ensured accurate scientific information delivery

SKILLS

Programming Languages	Python
Libraries	NumPy, Pandas, Scikit-Learn, Pytorch, Pyomo
Miscellaneous	L ^A T _E X,Git,SQL,PowerBI
Languages	Persian, English